



Time = 20 Minutes

Quiz No 05

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning.

CLO Assessed – CLO No 04 - Construct probability distributions, up to two dimensions, from given data for discrete and continuous random variables. (Cog - 3) – (applicable to both the questions.)

Question 1 [10 points] [Joint Distribution & Independence]

The joint probability distribution for the two Discrete Random Variables X and Y is shown below. The probability of the event $P(Y > X) = \frac{9}{50}$

Part a [5 points] Find the values for a & b.

Part b [5 points] Determine (with justification) whether random Variable X and Y are independent or not.

Y=	3	$\frac{1}{25}$	a	b
	2	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{5}{25}$
	1	$\frac{2}{25}$	$\frac{3}{25}$	$\frac{5}{25}$
		1	2	3
		X=		

Question 2 [10 points] [Function of a Random Variable & Expectation of a R.V.]

We have a tetrahedral die (4-sided) that has four outcomes, each side bearing numbers 1, 2, 3 and 4 with equal probabilities (fair die). We define X_1 and X_2 as the first and second outcome of the two independent rolls of the tetrahedral die, respectively. We define the following Discrete Random Variable:

$$W = (X_1 - X_2)^2$$

- (a) **[4 points]** Calculate and plot the Probability Mass Function, for the DRV, W .
- (b) **[6 points]** Calculate the Expected value, $E[W]$ and $\text{Var}[W]$.